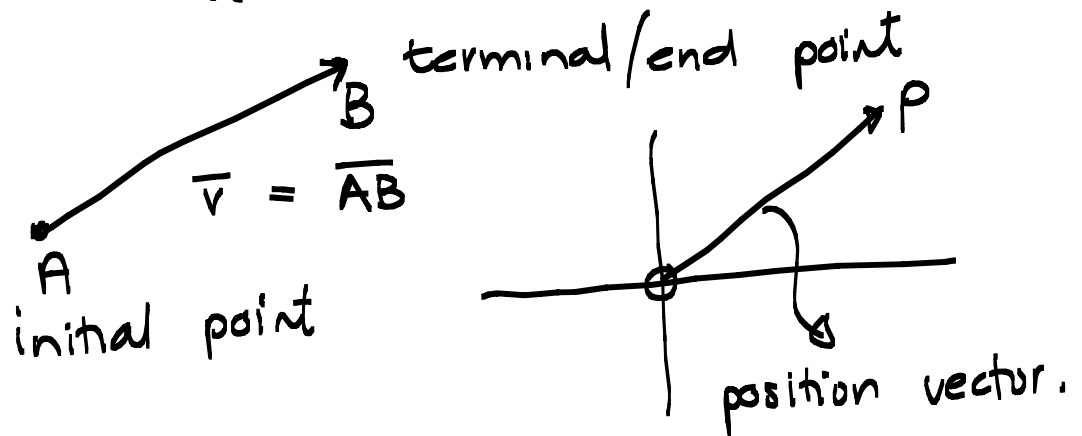
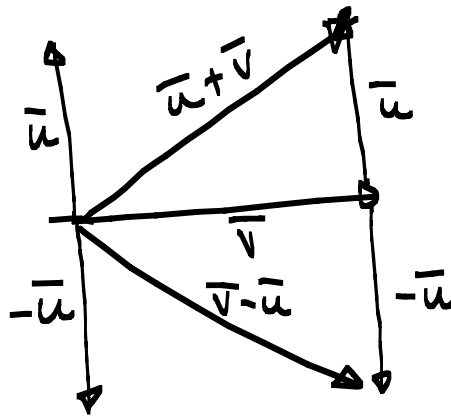


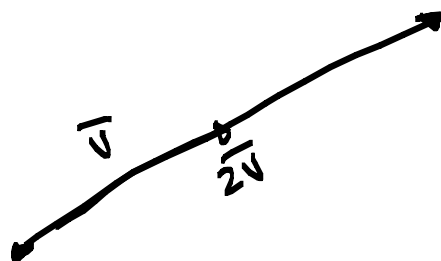
Vector - quantity with magnitude and direction.



Vector addition/subtraction



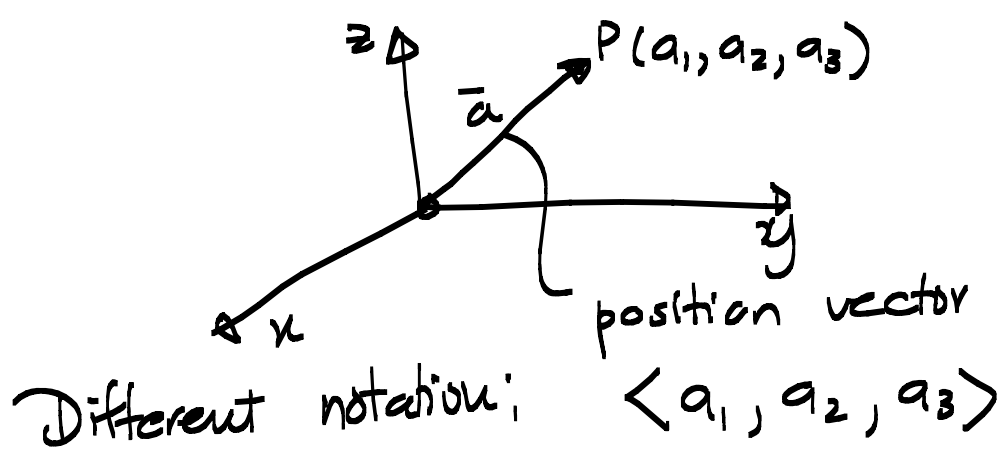
Scalar multiplication:



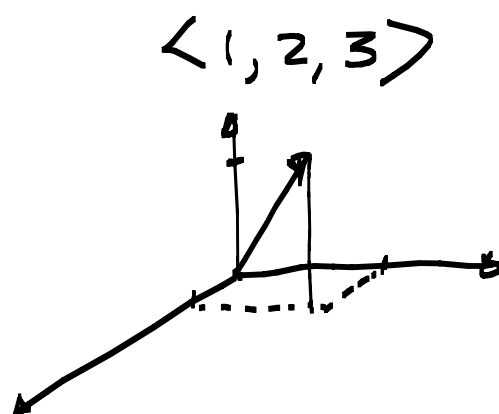
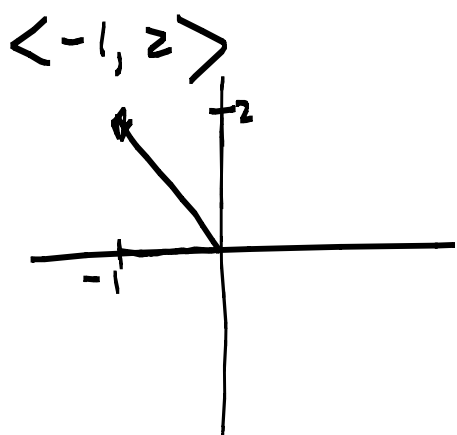
same direction
2x length

p 793 (SELF)

Components



Sketch:

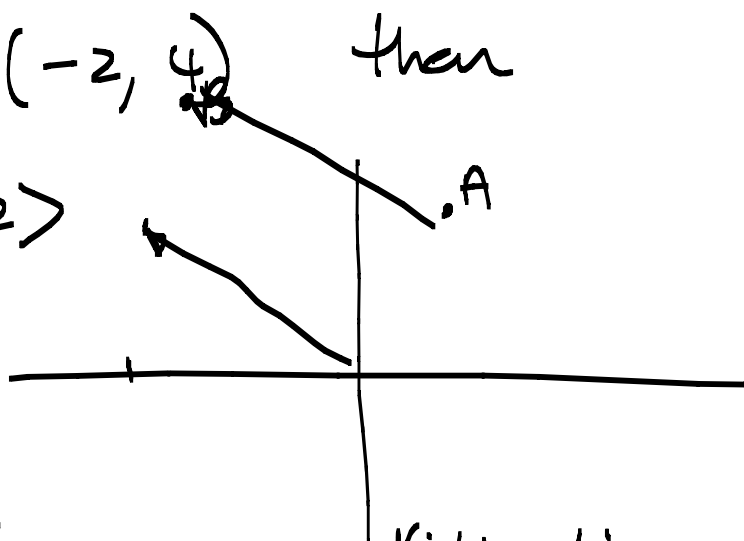


Given $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$
points in $\mathbb{R}^3$ then

$$\overline{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

If $A(1, 2)$ and $B(-2, 4)$ then

$$\overline{AB} = \langle -3, 2 \rangle$$



Addition / subtracted / scalar multiplication.

$$\bar{a} + \bar{b} = \langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$$

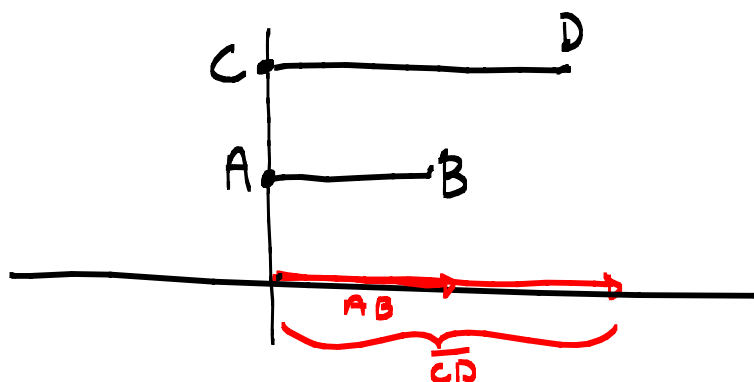
Find $2\bar{AB} - 3\bar{CA}$ if $\begin{matrix} A(2, -1, 0) \\ B(-3, 4, 1) \\ C(1, 2, -2) \end{matrix}$

$$\bar{AB} = \langle -5, 5, 1 \rangle \text{ and } \bar{CA} = \langle 1, -3, 2 \rangle$$

$$\Rightarrow 2\bar{AB} - 3\bar{CA} = \langle -13, 19, -4 \rangle$$

Vectors $\bar{a}$ and $\bar{b}$ are parallel. if there exists a $c \in \mathbb{R}$ so that $\bar{a} = c\bar{b}$

Example If $A(0, 1)$, $B(1, 1)$, $C(0, 2)$ and $D(3, 2)$ use a sketch and the definition to show that $\bar{AB} \parallel \bar{CD}$



$$\bar{AB} = \langle 1, 0 \rangle$$

$$\bar{CD} = \langle 3, 0 \rangle$$

$$\bar{AB} = \frac{1}{3}\bar{CD}$$

Magnitude / Length of a vector

In $\mathbb{R}^2$ $|\bar{a}| = \sqrt{a_1^2 + a_2^2}$ if $\bar{a} = \langle a_1, a_2 \rangle$

In $\mathbb{R}^3$ $|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ if $\bar{a} = \langle a_1, a_2, a_3 \rangle$

Unit vector has a length of 1.

If $\vec{a} \neq 0$ then $\frac{\vec{a}}{|\vec{a}|}$ is a unit vector in the direction of $\vec{a}$.

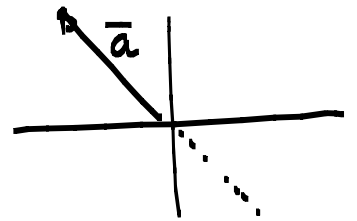
Example:

Find the vector of length 2 in the opposite direction as $\vec{a} = \langle -3, 4 \rangle$

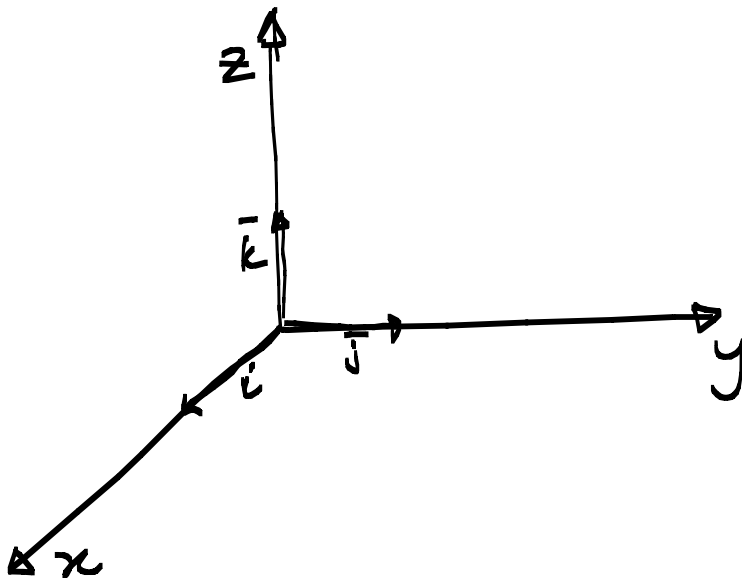
$$|\vec{a}| = \sqrt{9 + 16} = 5$$

Therefore the vector is

$$\begin{aligned} -2 \cdot \frac{|\vec{a}|}{5} &= -\frac{2}{5} \langle -3, 4 \rangle \\ &= \langle \frac{6}{5}, -\frac{8}{5} \rangle \end{aligned}$$



Standard basis vectors p796



$$\hat{i} = \langle 1, 0, 0 \rangle$$

$$\hat{j} = \langle 0, 1, 0 \rangle$$

$$\hat{k} = \langle 0, 0, 1 \rangle$$

$$* \langle 1, -2, 6 \rangle = 1\bar{i} - 2\bar{j} + 6\bar{k}$$

Find the unit vector in the direction
of $2\bar{i} - \bar{j} - 2\bar{k} = \langle 2, -1, -2 \rangle$

$$\text{length} = \sqrt{4+1+4} = 3$$

$$\begin{aligned} \bar{u} &= \frac{\langle 2, -1, -2 \rangle}{3} = \left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle \\ &= \frac{2}{3}\bar{i} - \frac{1}{3}\bar{j} - \frac{2}{3}\bar{k} \end{aligned}$$